MULTIVARIABLE CALCULUS HT23 SHEET 8
Gravity. Gauss' Flux Theorem. Poisson's Equation.

1. Two planets, modelled as particles of equal mass M, are situated at the points $(a,0,0)$ and $(-a,0,0)$ and remain there. Show that the gravitational potential at the point $(x,0,0)$, where $-a<x<a$, is

$$\frac{2GMa}{a^2-x^2},$$

and find the gravitational field at $(x,0,0)$. A space-craft moves along the segment $-a<x<a$ of the x-axis under the gravitational attraction of the planets. Write down the equation of motion for the space-craft and show that $x=0$ is a point of unstable equilibrium.

At $x=0$ the space-craft fires its motor briefly so as to acquire a speed $u(>0)$ and heads towards $x=a$. By using energy considerations show that the time taken to reach $x=a$ is

$$T=\int_0^a \sqrt{\frac{a^2-x^2}{u^2(a^2-x^2)+\frac{4GM}{a}x^2}}\,\mathrm{d}x.$$

(You are not required to evaluate the integral.)

2. A homogeneous straight wire of mass M lies along the x-axis from $(-a,0,0)$ to $(a,0,0)$. Show that the gravitational potential at the point $(x,y,0)$, where $y\neq 0$, is

$$\frac{GM}{2a}\left[\sinh^{-1}\left(\frac{a-x}{|y|}\right)+\sinh^{-1}\left(\frac{a+x}{|y|}\right)\right]$$

and that the x-component of the gravitational field is

$$\frac{GM}{2a}\left(\frac{1}{d_1}-\frac{1}{d_2}\right),$$

where d_1 and d_2 are the distances from $(x,y,0)$ to the ends of the wire.

3. Show that the gravitational field at the vertex of a homogeneous circular cone is

$$\frac{12GM}{a^2}\sin^2\frac{\alpha}{2},$$

where M is the mass of the cone, a is the radius of the base and α is the semi-vertical angle.

4. A hollow spherical shell has internal radius a and external radius b, and is made of material of uniform density ρ. Find the gravitational field and the gravitational potential in the three regions $0<r<a,\ a<r<b,\ r>b$ by using (1) the Flux Theorem, and (2) Poisson's equation.

5. A solid hemisphere of uniform density ρ occupies the region

$$x^2+y^2+z^2\leqslant a^2,\qquad z\leqslant 0.$$

Find the gravitational potential due to the hemisphere at the point $(0,0,s)$ where $s>0$. A uniform rod of density m per unit length lies on the z-axis between $(0,0,c)$ and $(0,0,d)$ where $d>c>0$. Show that the force exerted on the rod by the hemisphere is

$$\psi(c)-\psi(d)$$

where

$$\psi(\lambda)=\frac{2\pi Gm\rho}{3}\left(\frac{a^3+\lambda^3-(a^2+\lambda^2)^{3/2}}{\lambda}\right).$$